

Task 2: Understand the App and Build a Simple Model

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1 App Purpose and Main Features

This application demonstrates a basic local registration and login process. Users enter a username and password, which are stored in a local file called `credentials.txt`. During login, the app verifies the entered credentials against the stored data. If successful, a session token is generated and saved in `SharedPreferences`, and the user is redirected to the profile screen. Logging out clears the session token and ends the session.

2 Assumptions

We assume that the attacker can obtain the APK and reverse engineer it using standard static-analysis tools such as jadx. This allows the attacker to inspect the decompiled code, identify where security-sensitive values are generated, and understand how the application stores and uses credentials and session state. We also assume that the attacker can observe or approximate when a login event occurs, which helps narrow the time window in which a session token is generated.

3 Attacker Goals

The attacker's main goal is to compromise the app's authentication state by predicting, reproducing, or otherwise abusing the session token created after a successful login. In practical terms, the attacker aims to impersonate a legitimate logged-in user without following the intended authentication process. This threat model is directly relevant to our chosen vulnerability because the session token is generated using a non-cryptographic random source, making it a security-sensitive target.

4 Simple System Model

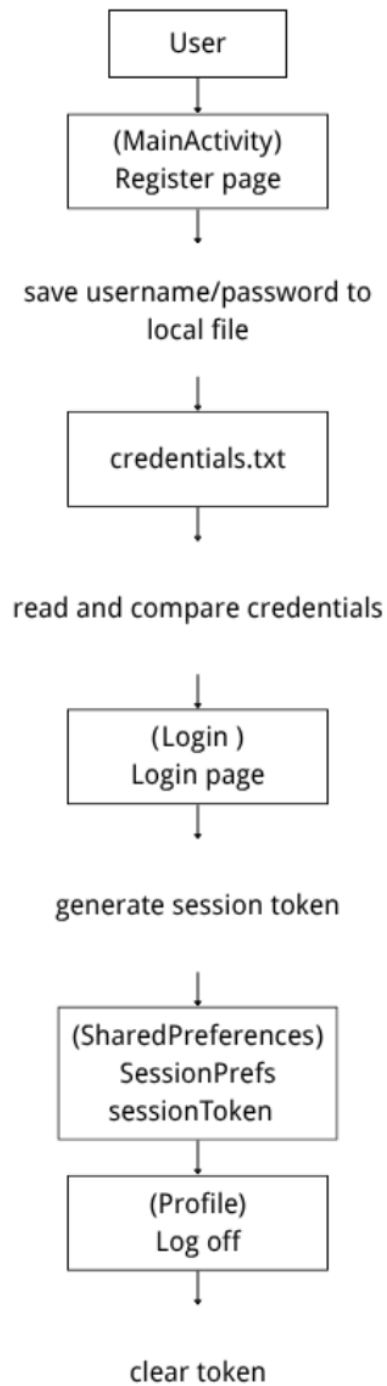


Figure 1: Simple system model